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MILITARY SPECIFICATION

CASE, PROJECTILE, BURSTER, M5

1. SCOPE

1.1 This specification covers Case, Projectile, Burster, M5.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-A-51	- Acetone, Technical
O-C-303	- Chromium Trioxide, Technical (Chromic Acid)
O-O-670	- Orthophosphoric (Phosphoric) Acid, Technical

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Federal

Federal Test Method Standard No. 141 - Paint, Varnish, Lacquer, and Materials; Method of Inspection, Sampling and Testing.

Federal Test Method Standard No. 151 - Metals; Test Methods

FSC 1315

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Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

F10535931	- Case, Projectile, Burster, M5
IEL-10535931	- Index of Inspection Equipment List for Case, Projectile, Burster, M5

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.1.1 Material.-The supplier shall be responsible for selecting from the classification list contained in the raw material specifications, the surface and physical condition of the raw material that will produce burster casings to the dimensions and mechanical properties specified as end item requirements.

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3.2 Stress relief.-When manufacturing the one piece design or alternative two piece design (casing and sleeve), the burster casing shall be stress relieved following the final forming operation. Components to be stress relieved must be brought up to a minimum temperature of 700°F and maintained at that temperature for a minimum of 30 minutes. For the two piece design, the casing only shall be stress relieved as specified above prior to assembly.

3.3 Mechanical properties.-Mechanical properties specified on the applicable drawing shall be applicable to all components unless otherwise specified.

3.4 Brazed joint (alternative designs).-The braze metal shall be visible for the full 360 degrees around the joint at the juncture of casing and rear of sleeve and also the juncture of the casing and base plug, outside, if applicable.

3.4.1 Joint sections (alternative designs).-When sectionalized, the braze metal joint shall be free of voids, cracks, entrapped slag, cleaning or other fluid for an aggregate of 85 percent of the exposed joint section.

3.5 Cleaning prior to testing.-The burster casing, prior to hydrostatic and air testing shall be free from rust and other foreign matter in order not to clog passages and possibly result in false non-leakers.

3.6 Protective coating.

3.6.1 Prior to application of the phenolic baking varnish, all surfaces of the burster case shall be thoroughly cleaned to remove all traces of grit, oil, wax, grease, rust or other foreign matter.

3.6.2 Final rinse.-The final rinse used in cleaning the burster case prior to application of the phenolic baking varnish, shall be maintained at a ph of 2.0 to 5.0 by the addition of a sufficient amount of chromic acid which shall comply with the requirements of O-C-303 (approximately .08 ounce per gallon of water) or a mixture of phosphoric acid, which shall comply with the requirements of O-O-670, and chromic acid (approximately .04 ounce of each per gallon of water). This rinse shall be maintained at a temperature of 140 degrees Fahrenheit to 210 degrees F.

3.6.3 Coating application.-The phenolic baking varnish shall be applied to the interior surfaces of the burster case within 4 hours after final cleaning operations have been completed. The varnish applied to the burster case shall produce an adherent film with no visible pinholes, craters, blisters or uneveness. Varnish shall be applied by a spray method unless otherwise authorized by the procuring activity.

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3.6.4 Cured coating.-After curing, the varnish coating shall be smooth and continuous and shall not soften or loosen when subjected to the acetone curing test. The acetone shall comply with the requirements of O-A-51.

3.6.5 Salt spray.-The cured varnish coating shall withstand exposure to the 20 percent salt spray for 24 hours without evidence of rust. However, rust within 1/32 inch of all edges shall be permitted.

3.7 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.-The burster case shall be free from cracks, splits, cold shuts, inclusions, porosity, laps, seams or other similar defects.

3.7.2 Burr.-The burster case shall contain no burr which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.-No burster case or assembly shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

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4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 10 burster cases as follows:

5 - with protective coating
5 - without protective coating

In addition to the above 10 burster cases, when alternative design is used, 5 sets of unassembled, uncoated, component parts shall be submitted.

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105, lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Tubing lots.-Components manufactured from tubing shall be fabricated from tubing lots containing not more than one heat of steel. The tubing lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.3 Bar stock lots.-Components manufactured from bar stock shall be fabricated from bar stock containing not more than one heat of steel. The bar stock lots thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

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4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with standard MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. When performing pressure tests, the items under test will be observed for noncompliance for the entire time it is required to be pressurized. All nonconforming material shall be rejected. The roughness comparison specimen prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects, a burr and foreign matter.

4.3.2.1 Tube, reinforcing - Drawing C-10535949 covering a detail of Drawing F10535931.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Inside diameter	Gage
102.	Thickness of wall	Gage
103.	Metal defective	Visual
MINOR		
201.	Min. flat of wall at front	Gage
202.	Min. flat of wall at rear	Gage
203.	Length total	Gage
204.	Width of slot, max.	Gage
205.	Radii or chamfers missing or incorrect	Visual (see 6.1)
206.	Surface finish, improper	Visual (see 6.1)
207.	Burr	Visual
208.	Foreign matter	Visual

4.3.2.2 Sleeve, (alternative two piece design) Drawing F10535948, covering a detail of Drawing F10535931.

CRITICAL		
1.	Metal defective	Visual

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Diameter of counterbore rear of sleeve	Gage
102. Diameter at rear of sleeve (outside)	Gage
103. Length, total	Gage
MINOR	
201. Length from rear of flange to bottom of counterbore rear of sleeve	Gage
202. Radii or chamfers missing or incorrect	Visual (see 6.1)
203. Surface finish improper	Visual (see 6.1)
204. Burr	Visual
205. Foreign matter	Visual

4.3.2.3 Casing, (alternative two piece and base design) Drawing F10534948 covering a detail of Drawing F10535931.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Diameter at front of tube (outside)	Gage
102. Diameter at rear of transition zone (outside)	Gage
103. Diameter of counterbore at rear (if applicable)	Gage
104. Depth of counterbore at rear (if applicable)	Gage
105. Thickness of wall at front of tube, min. (for min. indicated length)	Gage
106. Thickness of wall rear of transition zone	Gage
107. Thickness of wall at rear counterbore, min. (if applicable)	Gage
MINOR	
201. Radii or chamfers missing or incorrect	Visual (see 6.1)
202. Surface finish improper	Visual (see 6.1)
203. Burr	Visual
204. Foreign matter	Visual

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4.3.2.4 Plug (alternative base design) Drawing F10535948, covering a detail of Drawing F10535931.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Diameter, (small)	Gage
102. Length of small diameter	Gage
MINOR	
201. Radii or chamfers missing or incorrect	Visual (see 6.1)
202. Surface finish improper	Visual (see 6.1)
203. Burr	Visual
204. Foreign matter	Visual

4.3.2.5 Case, burster, Drawing F10535931 (prior to varnish).

CRITICAL	
1. Diameter of press fit surface	Gage
2. Presence of reverse taper in press fit surface	Gage
3. Profile of radius blend with press fit surface	Gage
4. Surface finish of press fit surface, improper (see 4.3.2)	Gage
5. Hydrostatic pressure stamp missing, (indicating failure to pass test) (if applicable)	Visual
6. Air pressure stamp missing, (indicating failure to pass test)	Visual
7. Braze metal not visible for full 360 degrees at exterior juncture of casing and rear of sleeve (if applicable)	Visual
8. Braze metal not visible for full 360 degrees at exterior juncture of casing and plug (if applicable)	Visual
9. Metal defective	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity of flange with press fit area	Gage
102. Concentricity at rear of tube with press fit area	Gage
103. Perpendicularity of front face of flange with press fit area	Gage
104. Diameter, of tube (rear of press fit surface)	Gage
105. Length of press fir surface min.	Gage
106. Length to end of taper at rear of press fit surface, min.	Gage
107. Location of reinforcing sleeve	Measure/Gage
108. Thickness through base	Gage
109. Thickness of wall (at rear of taper)	Gage
110. Improper cleaning	Visual
MINOR	
201. Diameter of flange	Gage
202. Length from rear of flange to bottom of cavity	Gage
203. Thickness of flange	Gage
204. Depth of undercut rear of flange, max. (if applicable)	Gage
205. Width of undercut rear of flange, max. (if applicable)	Gage
206. Reinforcing tube not properly brazed to body	Visual
207. Marking missing, incorrect or illegible	Visual
208. Burr	Visual
209. Foreign matter (except as otherwise classified)	Visual

4.3.2.6 Case, burster, Drawing F10535931 (after varnish).

CRITICAL	
None defined	
MAJOR	
101. Diameter over reinforcing tube	Gage
102. Protective coating, inadequate or damaged	Visual
103. Corrosion preventive compound missing from press fit surface	Visual
MINOR	
None defined	

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4.3.2.7 Carton (prior to sealing).

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Improper assembly	Visual
202. Incorrect number of cases in carton	Visual
203. Separator missing, or damaged	Visual

4.3.2.8 Carton (after sealing).

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual/Visual
202. Carton improperly sealed	Manual/Visual
203. Carton damaged, cut or gouged through all layers	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Hardness.

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4.3.3.1.1.1 Plug.-Ten plugs from each lot of plugs, when manufactured from sheet or strip steel shall be subjected to a hardness test with a Rockwell Hardness Tester to determine compliance with the applicable drawing. Failure of any plug to comply, shall be cause for rejection of the lot.

4.3.3.2 Tension test.

4.3.3.2.1 Tubing.-When the mechanical properties are not altered during manufacture, 2 tubes from each tubing lot, as defined by 4.3.1.2, shall be subjected to the tension test to determine compliance with the yield strength and elongation requirements of the applicable drawing. Failure of any specimen to comply shall be cause for rejection of the tubing lot.

4.3.3.2.2 Bar stock.-When the mechanical properties are not altered during manufacture, 2 bars from each bar stock lot, as defined by 4.3.1.3, shall be subjected to the tension test to determine compliance with the yield strength and elongation requirements of the applicable drawing. Failure of any specimen to comply shall be cause for rejection of the bar stock lot.

4.3.3.2.3 Burster case (one or two piece design).-Two burster casing, prior to insertion of the reinforcing tube, from each lot shall be subjected to the tension test to determine compliance with the yield strength and elongation requirements of the applicable drawing. Failure of any specimen to comply shall be cause for rejection of the lot.

4.3.3.3 Hydrostatic test.-(alternative designs) (critical defect) Each burster case, after cleaning but before gaging and varnishing, shall be subjected to this test. Failure of any burster case to comply with the requirements of the drawing shall be cause for rejection of the burster case. Burster cases which comply shall be stamped with the letter "H" as indicated on the applicable drawing.

4.3.3.4 Air test.-(critical defect) Each burster case shall be subjected to an internal air pressure test to determine compliance with the requirements specified on the applicable drawing. Failure to comply, as evidenced by the presence of bubbles on the surface of the burster case or rising to the surface of the water, shall be cause for rejection of the burster case. Burster cases that comply shall be stamped with the letter "A" as indicated on the applicable drawing.

4.3.3.5 Joint sections (alternative designs).-Five burster cases from each lot after hydrostatic test and air test, but prior to application of protective coating, shall be sectionalized to determine compliance with 3.4.1. Failure of any specimen to comply shall be cause for rejection of the lot.

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4.3.3.6 Protective coating.

4.3.3.6.1 Curing test.-One burster case from each 4 hour production or fraction thereof, shall be subjected to this test. Failure of the sample to comply with 3.6.4 shall be cause for rejection of the production represented by the sample.

4.3.3.6.2 Salt spray.-Two burster cases from each production lot shall be subjected to this test, failure of any burster case to comply with 3.6.5 shall be cause for rejection of the lot.

4.3.4 Straightness inside diameter (a major defect).-The inside diameter of each burster case, after protective coating, shall be gaged as prescribed by the applicable drawing. Failure of the functional gage to enter for the full indicated length shall be cause for rejection of the burster case.

4.3.5 Inspection equipment.-Index of Inspection Equipment Lists IEL10535931 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL.

4.4 Test methods.

4.4.1 Mechanical properties.

4.4.1.1 Hardness.

4.4.1.1.1 Plug.-The samples selected in accordance with 4.3.3.1.1 shall be tested using the equipment and procedures prescribed for this test by Federal Test Method Standard No. 151. The test may be made on either face of the plug.

4.4.1.2 Tension test.

4.4.1.2.1 Tubing.-From the samples selected in accordance with 4.3.3.2.1, one full section tension test specimen as prescribed by Federal Test Method Standard No. 151 shall be taken from each of the 2 tubes. The test shall be performed using equipment and procedures prescribed by Federal Test Method Standard No. 151.

4.4.1.2.2 Bar stock.-From the samples selected in accordance with 4.3.3.2.2, one longitudinal tension test specimen conforming to the largest obtainable standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken from each of 2 bars. The test shall be performed using equipment and procedures prescribed by Federal Test Method Standard No. 151.

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4.4.1.2.3 Burster case (one or two piece design).-From the samples selected in accordance with 4.3.3.2.3, one full section tension test specimen as prescribed by Federal Test Method Standard No. 151 shall be taken from the tube portion of each of the 2 burster casings. The test shall be performed using equipment and procedures prescribed by Federal Test Method Standard No. 151.

4.4.2 Hydrostatic test.-(alternative design) The burster case shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. The burster case shall be mounted in the test fixture and held in place by clamping the sleeve flange; no other portion of the burster case will be restrained during application of pressure. The pressure will be applied until the minimum pressure is reached and then held for the minimum time specified on the applicable drawing. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours during operation. It shall also be calibrated prior to re-starting the operation after a stoppage of 2 hours or more.

4.4.3 Air test.-Each burster case shall be mounted in an approved fixture in such a manner that the burster case will be completely submerged in water. When performing this test, the inspector will ascertain that the required pressure is reached and the water has been permitted to settle after immersion of the case undertest prior to start of timing; observation for leakage shall be made for the entire time the case is required to be pressurized. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours during operation. It shall also be calibrated prior to re-starting the operation after a stoppage of 2 hours or more.

4.4.4 Joint section (alternative design).-The samples selected in accordance with 4.3.3.4 shall be sectionalized into 2 segments by cutting longitudinally along the assembly axis. Care shall be exercised to assure that no damage occurs to the brazed joint section. The braze metal of the joints of each segment shall be visually examined to determine compliance with 3.4.1.

4.4.5 Protective coating.

4.4.5.1 Curing test.-An area of at least 1 square inch on the sample burster case, shall be kept wet with acetone for 2 minutes. The area being tested shall be vigorously rubbed with the thumb and visually inspected for evidence of lifting, blistering, or softening of the varnish coating.

4.4.5.2 Salt spray test.-The procedures for testing the cleaning and application of protective coating shall be in accordance with the applicable specifications. Salt spray test shall be performed using equipment and procedures prescribed by Federal Test Method Standard No. 141.

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4.4.6 Inside diameter.-The functional gage shall be entered into the burster case to the length indicated on the applicable drawing.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level C - Thrity-six (36) preserved burster cases (without further overwrapping) shall be packed vertically in a 6 by 6 arrangement in separate cells in accordance with MIL-STD-1169. Full length half-slotted partitions shall provide complete separation between items.

5.2 Marking.-Marking shall be in accordance with MIL-STD-1169.

5.3 Data card.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Protective coating.

6.2.1 Application.-Care should be taken in the application of the varnish to prevent formation of uncoated areas on the interior due to vapor wash or other causes.

6.2.2 Color.-Slight variations of color of the coated surface due to chromic acid rinse shall not be cause for rejection.

6.2.3 Rinsing.-Adequate rinsing is important in the cleaning process. When the water film on the burster case, after the cleaning operation maintains its continuity (is free from water break), the surface may be considered free from oil and grease.

6.3 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

Custodian

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Preparing activity

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Project No. 1315-A723

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